

Operations Research

Final Project 2026

A mid-sized airline operates a set of round-trip international flights from its base at **Dubai International Airport (DXB)**. Each round trip consists of an outbound flight from DXB to a destination and a return flight on the same or next day.

Your task is to **design an optimal crew scheduling model** to assign pilots and cabin crew to each round-trip pair of flights. You must satisfy operational requirements, adhere to labor regulations, and minimize the overall cost of crew assignment.

You are provided with the following datasets:

- `flight_pairs.csv`: round-trip flight pairs over a month.
- `crew.csv`: list of available pilots and cabin crew, with qualifications and pay

You must consider the following constraints in your model:

1. Coverage Constraints: each flight pair must be assigned 1 pilot and 2 cabin crew members.
2. Crew Qualification: a crew member can only be assigned to a flight pair if they are qualified to operate the aircraft type used in that pair.
3. Maximum Workload: each crew member has a maximum number of flight pairs they can be assigned per month. Additionally, pilots should not fly for more 100 hours per month, and cabin crew members shouldn't fly for more than 150 hours per month.
4. Minimum Rest Time Between Pairs: If a pilot or cabin crew member is assigned to two different flight pairs, there must be at least 24 hours of rest between the arrival time of the first pair and the departure of the second pair.

Questions:

1. Find the optimal pilot and cabin crew assignment to minimize the monthly operational costs (the budget assigned to pilots' salaries and to the budget assigned to cabin crew salaries).
2. If regulations have changed **for pilots only**, and the minimum rest duration between flights have become 48 hours (instead of 24), and the maximum flying hours have become 50 hours per month (instead of 100), how would that impact the budget assigned to pilots' salaries?
3. If the company wants to keep the budget assigned to pilots' salaries to what was obtained in (1). Hence, they cannot operate all the flight and have to cancel some flights. Which flights should the company operate or cancel such that they maximize their revenue? How much percentage of the maximum revenue which the company can get from all flight would it lose if they cancel these flights? (hint: use the `EstRevenue` column in the `flight_pairs` dataframe).
4. Without solving, explain how would you include each crew member vacations (if given as additional column in the crew data) in your model.

Hint: you can set the stopping criteria for the integer programming solver to 120 seconds and 1% optimality gap as follows:

```
solver = SolverFactory("cbc", executable=cbc_path)
solver.options["ratioGAP"] = 0.001
solver.options["seconds"] = 120
```

Guidelines:

- *Deliver*: A working code notebook & 2-page max pdf file explaining your model, solution and the results.
- *Teams*: up to 3 members per team.
- *Deadline*: Sunday 12-April-2026.